



Spotify Taste Analysis

Total Songs

195

Average Energy

0.64

Average Danceability

0.64

Average Tempo

121.09

Average Speechiness

0.15

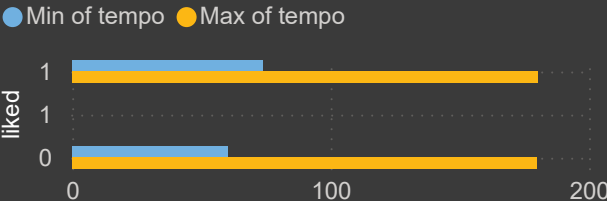
Average Liveness

0.15

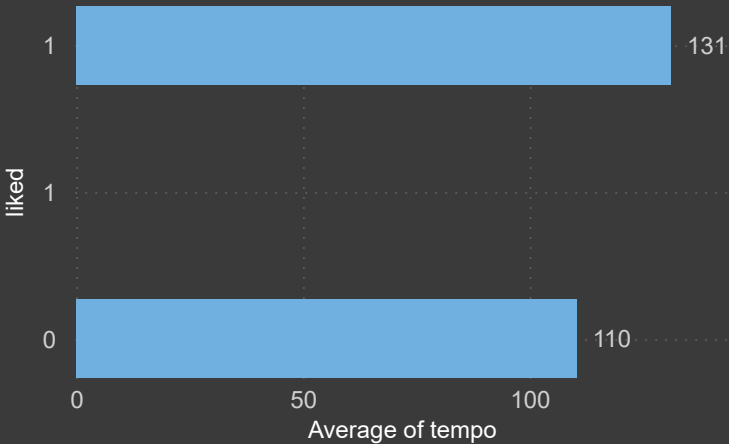
Average Valence

0.49

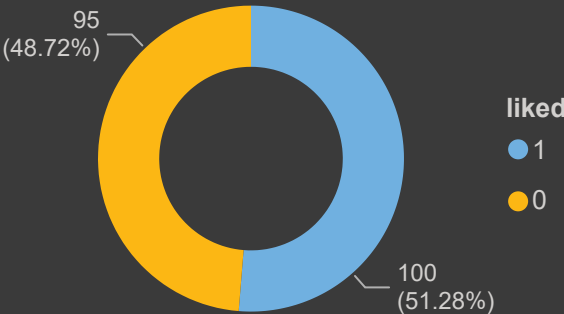
Tempo by liked



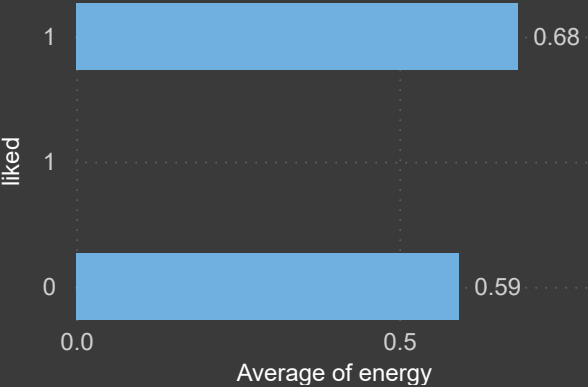
Average Tempo By Like Statue



Liked vs Not Liked songs



Average Energy by Like Statue

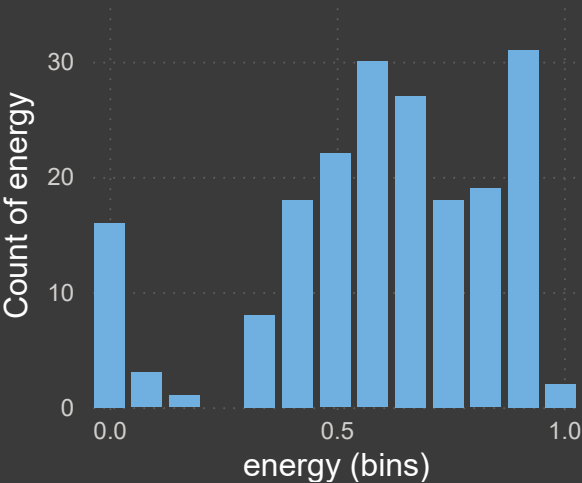




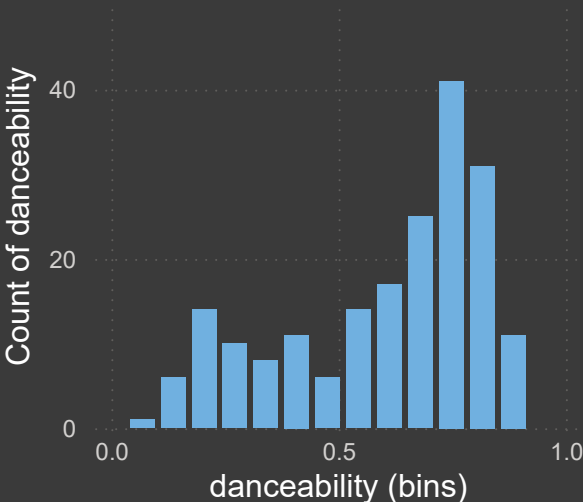
Song Characteristics



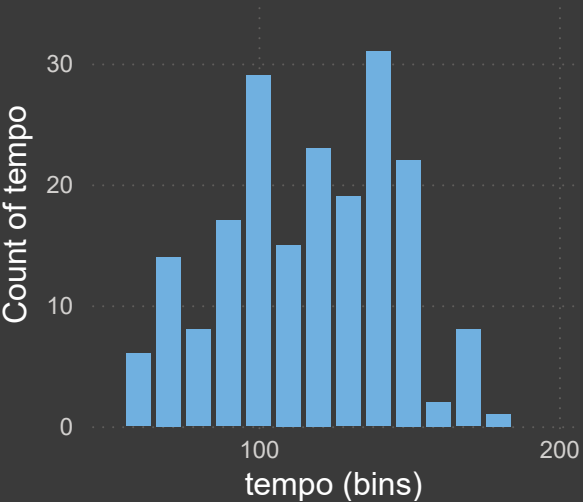
Energy Distribution



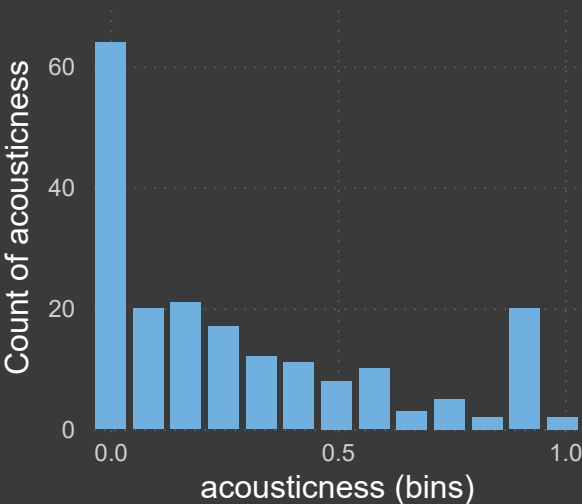
Danceability Distribution



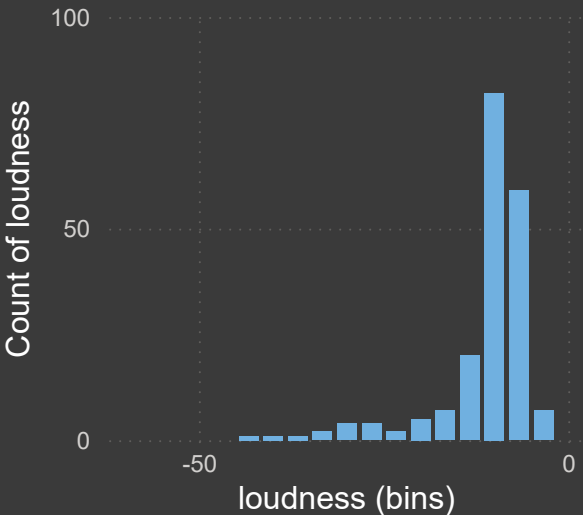
Tempo Distribution



Acousticness Distribution

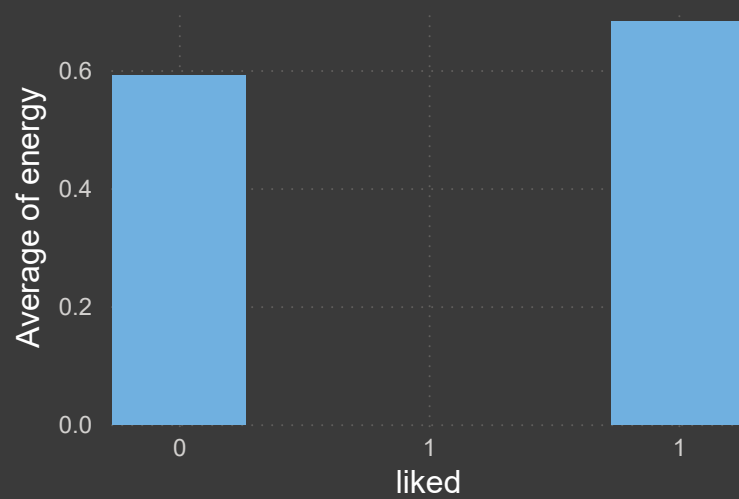


Loudness Distribution

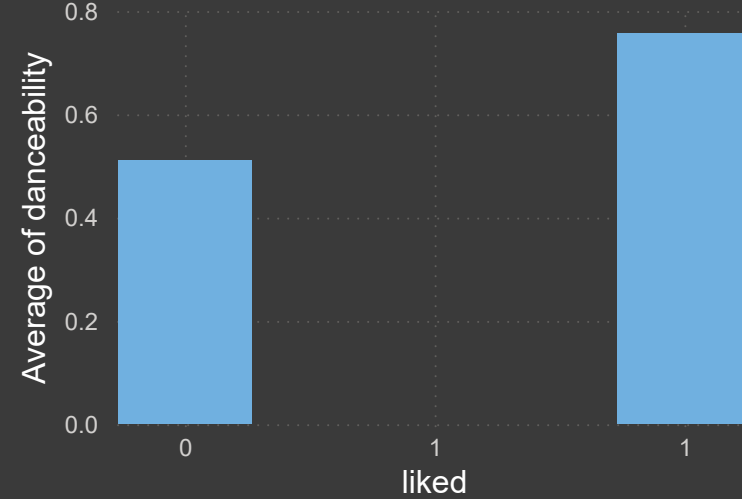


Liked Songs Analysis

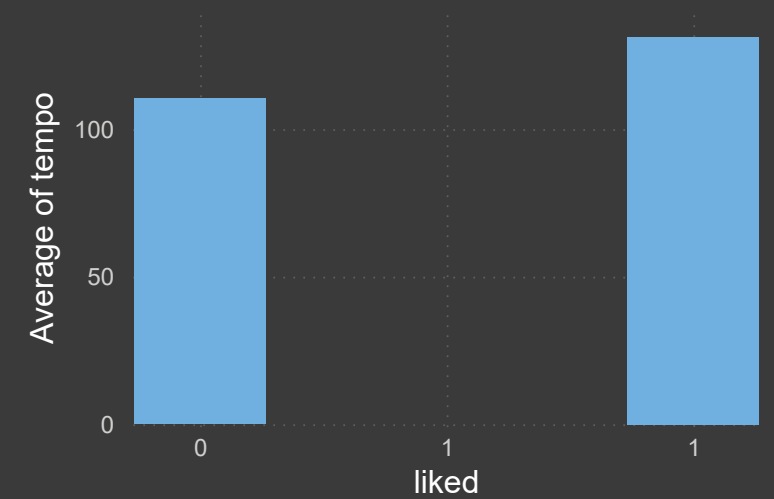
Energy Comparison



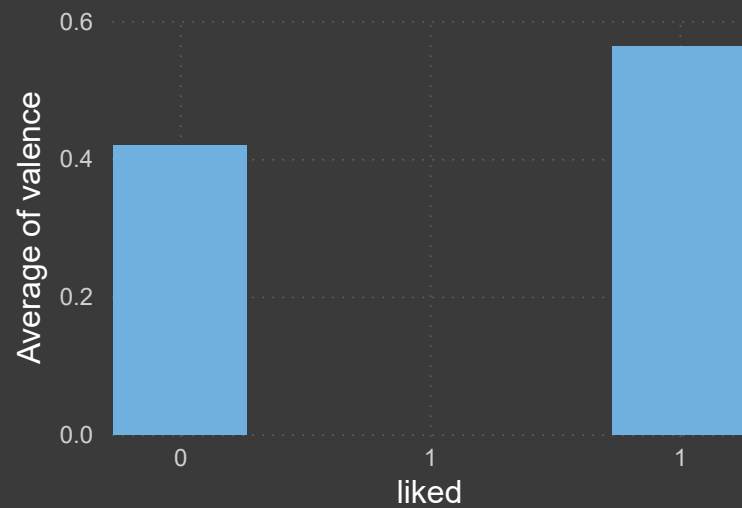
Danceability Comparison



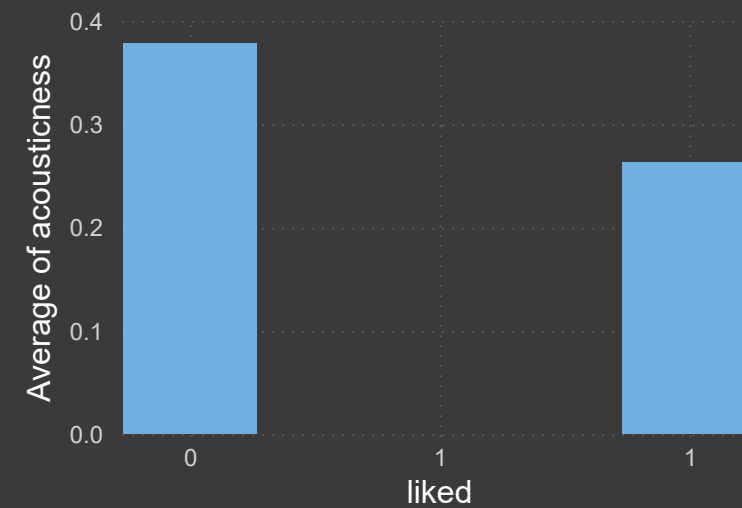
Tempo Comparison



Valence Comparison



Acousticness Comparison



Insights & Recommendations

Data-driven analysis of song characteristics and listener preferences

Key Metrics	Total Songs	Liked Songs	Liked Songs %	Avg Danceability of Liked
	195	100	51.28%	75.66%

Key Findings

- Liked songs show higher energy:** The average energy of liked songs is 0.68, compared with 0.59 for non-liked songs.
- Danceability is noticeably higher in liked songs:** Liked songs have an average danceability of 0.72, compared with 0.50 for non-liked songs.
- Liked songs have a higher average tempo:** The average tempo is approximately 127 BPM for liked songs compared with approximately 110 BPM for non-liked songs.
- Liked songs have higher valence:** The average valence of liked songs is approximately 0.56, compared with approximately 0.42 for non-liked songs.
- Liked songs have lower acousticness:** Average acousticness is approximately 0.25 for liked songs compared with approximately 0.37 for non-liked songs.

Business Recommendations

- Personalized Playlists:** Use song characteristics such as danceability, energy, tempo and valence as additional signals when creating personalized playlists.
- Preference-Based Segmentation:** Group songs according to their audio characteristics and identify which groups receive higher user preference.
- Recommendation Strategy:** Give greater consideration to tracks with characteristics that show stronger association with the liked category.
- Playlist Development:** Create different playlists based on listener preference profiles, such as high-energy, highly danceable or positive-valence tracks.
- Improve Future Analysis:** Combine audio features with user behaviour such as skips, listening duration, repeat plays and playlist additions.

What the Analysis Suggests

The analysis suggests that songs with higher danceability, energy, tempo and valence are more strongly associated with the liked category in this dataset.

Liked songs also show lower average acousticness compared with non-liked songs.

These patterns can help identify song characteristics that are associated with listener preferences.

Analysis Limitations

- The dataset contains 195 songs and may not represent the preferences of all Spotify users.
- The analysis identifies association between audio features and the liked category; it does not establish causation.
- The current dataset does not include detailed user behavior such as skips, listening duration or repeat plays.
- A larger dataset with user-level behavior could provide stronger insights and improve recommendation analysis.

Conclusion

The Spotify Taste Analysis project examined 195 songs using audio characteristics such as energy, danceability, tempo, valence and acousticness. The analysis shows clear differences between liked and non-liked songs, with liked songs displaying higher average energy, danceability, tempo and valence, while showing lower acousticness. These findings can support data-driven playlist personalization and recommendation strategies.